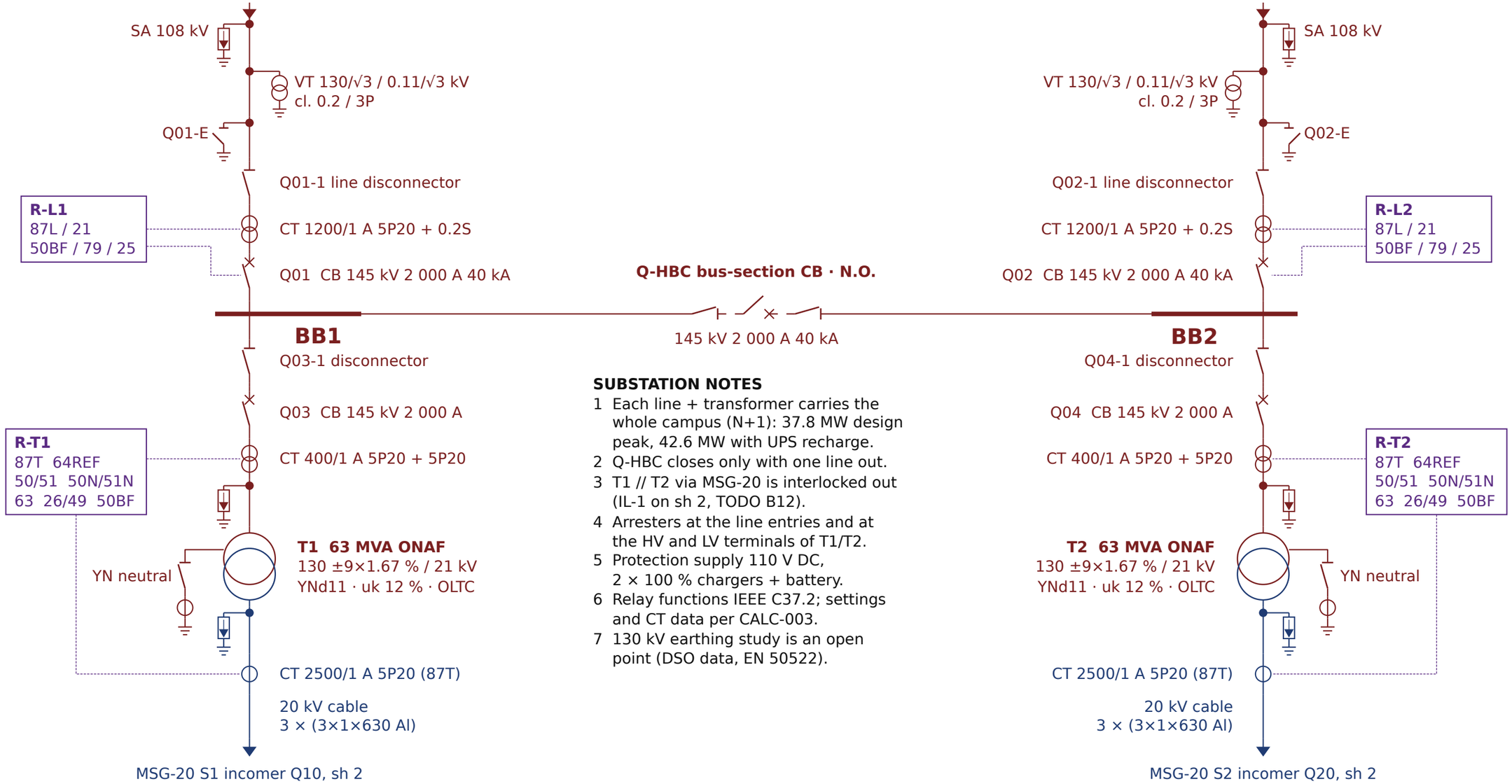

SUBSTATION NOTES

1 Each line + transformer carries the whole campus (N+1): 37.8 MW design peak, 42.6 MW with UPS recharge.
2 Q-HBC closes only with one line out.
3 T1 // T2 via MSG-20 is interlocked out (IL-1 on sh 2, TODO B12).
4 Arresters at the line entries and at the HV and LV terminals of T1/T2.
5 Protection supply 110 V DC, 2 × 100 % chargers + battery.
6 Relay functions IEEE C37.2; settings and CT data per CALC-003.
7 130 kV earthing study is an open point (DSO data, EN 50522).

Short-circuit level (CALC-003/004)	Value
130 kV S''k max / min (A)	3 000 / 1 500 MVA
20 kV normal (S0)	13.0 kA, ip 34.4 kA
20 kV with BESS	13.3 kA
Worst abnormal + BESS	23.1 kA = 92 % of 25
T1 // T2 (not permitted)	ip 58.5 kA = 94 %

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
Portfolio D. Wu · public standards and [ASSUMED] data only

130/20 kV substation — sheet 1

H-arrangement, T1/T2 to the MSG-20 incomers

CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-001	**1 / 2**	**P2**